

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Civil and Environmental Engineering) SEMESTER - 4 Winter 2025 (Supply.)

Course :Bachelor of Technology (Civil and Environmental Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 4

Subject Code & Name: BTCEC402 - ENVIRONMENTAL CHEMISTRY AND MICROBIOLOGY

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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- Q1. Objective type questions. (Compulsory Question) 12
- 1 Which law states that mass is neither created nor destroyed during a chemical reaction?
 - a) Law of definite proportions
 - b) Law of multiple proportions
 - c) Law of conservation of mass
 - d) Avogadro's law
 - 2 Colloidal particles range in size approximately between:
 - a) <1 nm
 - b) 1–1000 nm
 - c) >1000 nm
 - d) >1 mm
 - 3 DDT belongs to which category of pesticide?
 - a) Organophosphate
 - b) Carbamate
 - c) Organochlorine
 - d) Pyrethroid
 - 4 Which detergent type is most biodegradable?
 - a) Branched alkyl benzene sulfonate
 - b) Linear alkyl sulfonate
 - c) Cationic detergent
 - d) Non-ionic detergent
 - 5 Which compound is most harmful to aquatic microorganisms?
 - a) Biodegradable surfactant
 - b) Persistent organochlorine pesticide
 - c) Short-chain alcohol
 - d) Dissolved oxygen
 - 6 Beer–Lambert law relates absorbance with:
 - a) Temperature
 - b) Pressure
 - c) Concentration
 - d) Density

- 7 Gas chromatography is mainly used for analysis of:
 - a) Non-volatile compounds
 - b) Suspended solids
 - c) Colloids
 - d) Volatile organic compounds
- 8 Which technique is best for pesticide residue analysis?
 - a) Spectrophotometry
 - b) Fluorometry
 - c) Gas chromatography
 - d) Microscopy
- 9 Which staining differentiates Gram-positive and Gram-negative bacteria?
 - a) Simple staining
 - b) Acid-fast staining
 - c) Gram staining
 - d) Capsule staining
- 10 Absence of oxygen in sludge digesters promotes:
 - a) Aerobic bacteria
 - b) Anaerobic bacteria
 - c) Algae
 - d) Fungi only
- 11 Enzymes are chemically:
 - a) Carbohydrates
 - b) Lipids
 - c) Proteins
 - d) Vitamins
- 12 Micrometry deals with:
 - a) Identification
 - b) Measurement
 - c) Culturing
 - d) Staining

Q2. Solve the following.

- A) Explain the role of **acid–base equilibrium** in controlling water quality of natural water bodies with suitable examples. 6
- B) Apply principles of **oxidation–reduction reactions** in understanding corrosion and its environmental implications. 6

Q3. Solve the following.

- A) Explain the chemistry and environmental behavior of **DDT** with reference to persistence and bioaccumulation. 6
- B) Explain pesticide degradation mechanisms with suitable environmental examples. 6

Q4. Solve Any Two of the following.

- A) Explain the application of **flame photometry** for estimation of alkali metals in water samples. 6
- B) Evaluate the effectiveness of **ion chromatography** for water quality assessment. 6
- C) Describe the working principle of **fluorometry** and its environmental applications. 6

Q5. Solve Any Two of the following.

- A) Apply microbiological principles to explain **aerobic wastewater treatment processes**. 6
 - B) Analyze the differences between **aerobic and anaerobic microbial processes**. 6
 - C) Describe microbial respiration pathways relevant to environmental engineering. 6
- Q6. Solve Any Two of the following.
- A) Explain different **staining techniques** used in environmental microbiology with applications. 6
 - B) Analyze the significance of **MPN and plate count methods** in water quality monitoring. 6
 - C) Assess the importance of industrial microbiology in environmental engineering applications. 6

*** End ***